

YenCare Health Platform

Final Week Master Sprint Plan & Capstone Defense Roadmap (Gate 5)

Programme / Context	AmaliTech CSR Capstone Internship Programme — Product 4 (KNUST Health Services)
Active Sprint State	Gate 4 Hardening Verified • Final Week Gate 5 Capstone Defense & Handover
Sprint Execution Period	Monday, 28 September 2026 – Friday, 2 October 2026 (Final Capstone Week)
Final Week Objective	Close code hygiene gaps (ESLint, NoSQL sanitizer), verify cloud staging host, execute live multi-role simulation rehearsal, deliver 15-minute capstone defense, and package repository bundle v1.0.0.
Automated Test Baseline	391 Passing Tests across 94 Suites (100% Clean native Node Test Runner)
End-to-End Test Baseline	37 Passing, 11 Skipped, 0 Failing via Playwright in CI
Production Build Status	100% Clean Vite Production Bundles for both frontend and ui/prototype
Lead Author & Role	Blessing Edmund Kwame Dogbe (Backend & DevOps Lead / Product Coordinator)

1. Final Week Strategic Approach & Phase Roadmap

With all core clinical features, security policies, and reception workflows hardened and verified in Week 4, the engineering team enters the final week with zero blocking engineering defects. Work is organized into three sequential phases:

Phase & Timeframe	Target Focus	Core Deliverables & Outcomes
Phase 1 (Mon – Tue)	Code Hygiene & Cloud Staging	Resolve 12 frontend ESLint issues, mount NoSQL operator sanitizer, verify Render cloud staging backend and MongoDB Atlas connectivity.
Phase 2 (Wed – Thu)	Simulation & Presentation Prep	Conduct multi-role live simulation rehearsal (P01-S10), model clinic throughput impact (65% congestion reduction), record 3-min 1080p demo video, build pitch deck.
Phase 3 (Friday)	Capstone Defense & Handover	Deliver 15-minute presentation to AmaliTech evaluation board, conduct live demonstration, tag v1.0.0 release, and deliver handover package.

Gate 5 Milestone: Capstone Defense & Long-Term Institutional Value

- YenCare directly addresses outpatient overcrowding and lost student academic hours at KNUST Health Services.
- The final week ensures that technical excellence (391 tests, zero security bypasses, fail-fast env validation) is paired with flawless stakeholder presentation and turn-key operational handover.
- All team members have defined ownership across presentation delivery, live demonstration roles, and technical documentation.

2. Prioritized Remaining Tasks & Engineering Cards

[SEC-NOSQL] Defense-in-Depth Generic NoSQL Operator Sanitizer	
Priority & Milestone	PRIORITY 1 (SECURITY) Final Week (Gate 5)
Assignee & Support	Primary: Emmanuella Lodonu (Backend & Security) • Support: Able Kafu Azanda (Backend Lead)
Assigned Scope	System-wide HTTP API Security Layer (backend/src/http/security.js, app.js)
Rationale	Augments existing route-level parameter validation with global defense-in-depth protection against nested NoSQL query operator injections.
Scope & Goals	Mount generic recursive Express middleware in app.js that sanitizes all incoming request bodies, query strings, and URL parameters by stripping or escaping keys beginning with '\$' or containing '.', preventing NoSQL operator injection attempts.
Dependencies	Blocked by: None • Blocks: Production security compliance sign-off. • Target: Monday, 28 Sep 2026, 5:00 PM
Acceptance Criteria	- Express middleware mounted before all application route handlers. - Sanitizes or rejects payloads containing keys like '\$gt', '\$ne', '\$regex', or dotted properties. - Automated unit tests added in backend/test/ verifying operator stripping. - Zero regressions across the existing 391 passing backend tests.

[FE-LINT] Frontend Code Hygiene & React 19 ESLint Clean-Up	
Priority & Milestone	PRIORITY 2 (QUALITY & HYGIENE) Final Week (Gate 5)
Assignee & Support	Primary: Raymond B. Afrani (Frontend Lead) • Support: Harry “Ephraim” Nartey (Fullstack)
Assigned Scope	frontend/src/pages/ (ClinicActivity, StaffPortal), frontend/src/components/ (ReviewBooking, WelcomeForm, ElapsedTimer)
Rationale	Resolves 12 ESLint problems (8 errors, 4 warnings) identified in audit and integrates linting into the CI pipeline.
Scope & Goals	1. Remove unused Button import in ReviewBooking.jsx. 2. Refactor synchronous setState inside useEffect bodies in WelcomeForm.jsx, ElapsedTimer.jsx, useClinicRoster.js, and ClinicActivity.jsx. 3. Separate helper exports from component files in StaffAuthContext.jsx and ElapsedTimer.jsx. 4. Add frontend linting step to .github/workflows/pr-ci.yml.
Dependencies	Blocked by: None • Blocks: Pristine CI gate enforcement. • Target: Tuesday, 29 Sep 2026, 5:00 PM
Acceptance Criteria	- npm --prefix frontend run lint completes with 0 errors and 0 warnings. - All components render and state transitions remain functional. - GitHub Actions PR CI workflow runs frontend lint step on every push.

[OPS-HOST] Render Staging Backend Verification & Cloud DNS Uptime	
Priority & Milestone	PRIORITY 3 (DEVOPS & STAGING) Final Week (Gate 5)
Assignee & Support	Primary: Blessing Edmund Kwame Dogbe (DevOps / Lead) • Support: AmaliTech Cloud Infrastructure Support
Assigned Scope	Production Cloud Infrastructure (Render yencare-api-staging, Vercel yencare-platform, MongoDB Atlas)
Rationale	Ensures cloud staging environment is active, reachable, and correctly paired with MongoDB Atlas and the live Vercel frontend.
Scope & Goals	Verify and un-pause the Render web service for yencare-api-staging.onrender.com. Ensure MONGODB_URI, JWT_SECRET, and MNOTIFY_API_KEY environment variables match production requirements. Verify public GET /health responds with 200 OK and test cross-origin requests from Vercel.
Dependencies	Blocked by: Render dashboard access • Blocks: Live clinical rehearsal on cloud. • Target: Tuesday, 29 Sep 2026, 5:00 PM
Acceptance Criteria	- GET https://yencare-api-staging.onrender.com/health returns 200 OK with db: 'connected'. - CORS allows cross-origin requests from https://yencare-platform.vercel.app. - End-to-end booking from live Vercel app writes to MongoDB Atlas and dispatches SMS.

[QA-SIM] Live Multi-Role Clinical Simulation & Rehearsal	
Priority & Milestone	PRIORITY 4 (CLINICAL QA) Final Week (Gate 5)
Assignee & Support	Primary: Sterling Awuley (QA Lead) + All Team Members • Support: Blessing Edmund Kwame Dogbe
Assigned Scope	P01-P07 (Intake), P09-P17 (Self-Service), S01-S03 (Reception Roster), S04-S05 (Doctor Room), P18/S10 (Corridor Display)
Rationale	Validates seamless multi-role handoff between student mobile client, receptionist desk, doctor workstation, and corridor display.
Scope & Goals	Execute the comprehensive 5-journey testing runbook from docs/testing/LIVE_TESTING_GUIDE.md: 1. Student online booking. 2. Patient self-service OTP reschedule. 3. Arrival check-in within window. 4. Receptionist roster check-in & walk-in triage. 5. Doctor workstation room isolation and consultation completion. 6. Corridor live queue screen updates.
Dependencies	Blocked by: [OPS-HOST] staging backend verification • Blocks: Final presentation demonstration. • Target: Wednesday, 30 Sep 2026, 5:00 PM
Acceptance Criteria	- All 5 clinical journeys completed without uncaught exceptions or UI dead-ends. - Doctor room isolation verified (Dr. Kwame in Room 1 never sees Dr. Ama's patients in Room 2). - Real SMS delivered to Ghana phone numbers or logged in staging logs. - Complete manual runbook execution logged with evidence.

[SLIDES-DEMO] Stakeholder Pitch Deck, Impact Modeling & Demo Video	
Priority & Milestone	PRIORITY 5 (PRESENTATION) Final Week (Gate 5)
Assignee & Support	Primary: Raymond B. Afrani & Harry "Ephraim" Nartey • Support: Blessing Edmund Kwame Dogbe
Assigned Scope	Executive Presentation Materials, System Architecture Diagrams, Clinical Impact Analytics
Rationale	Delivers a compelling, data-backed 15-minute presentation and offline contingency video for the AmaliTech Evaluation Board.
Scope & Goals	1. Build a 15-minute presentation slide deck covering the KNUST healthcare problem, YenCare solution, architecture, security, and live demo. 2. Model simulated clinic throughput showing up to 65% reduction in waiting room peak congestion. 3. Record a 3-minute 1080p offline demo video showcasing the complete clinical flow as a backup for network outages.
Dependencies	Blocked by: [QA-SIM] simulation completion • Blocks: Gate 5 presentation. • Target: Thursday, 1 Oct 2026, 5:00 PM
Acceptance Criteria	- Final pitch deck approved by team lead and mentors. 3-minute high-definition video walkthrough recorded and archived locally. - Timed rehearsal completed within the 15-minute presentation window.

[HANDOVER] Final Repository Bundling, Documentation & Stakeholder Handover	
Priority & Milestone	PRIORITY 6 (HANDOVER) Final Week (Gate 5)
Assignee & Support	Primary: Blessing Edmund Kwame Dogbe (Lead) • Support: Able Kafu Azanda (Backend Lead)
Assigned Scope	docs/ Runbooks (api, architecture, operations, security, testing), .env.example, GitHub Release v1.0.0
Rationale	Ensures the YenCare platform is cleanly packaged, documented, and archived for long-term deployment by KNUST Health Services.
Scope & Goals	1. Synchronize all environment templates and documentation in docs/. 2. Verify clean git clone, seed, and build from scratch. 3. Tag GitHub release v1.0.0 with comprehensive changelog and release notes. 4. Prepare administrative handover dossier with credentials, maintenance commands, and architecture overview.
Dependencies	Blocked by: All preceding tasks • Blocks: Final internship capstone sign-off. • Target: Friday, 2 Oct 2026, 12:00 PM (Defense at 2:00 PM)
Acceptance Criteria	- Clean installation verified from a fresh clone. - GitHub release v1.0.0 tagged on main with release notes. - All documentation runbooks updated and exported as PDFs. - Formal handover dossier signed off and delivered to evaluators.

3. Final Quality Verification Gates & Test Standards

Verification Layer	Tooling & Command	Target Threshold	Current Status
Backend Unit & Integration	npm run test:backend	391+ tests, 0 failures	391 Passing (100%)
End-to-End Automation	npm run test:e2e	37+ passing, 0 failures	37 Passing in CI
Production Client Build	npm run build:frontend	0 bundle errors	Built in 16.4s
Prototype Client Build	npm run build	0 bundle errors	Built in 1.1s
Frontend Code Hygiene	npm run lint	0 errors, 0 warnings	12 issues to fix
Staging Cloud Health	GET /health	200 OK (db: 'connected')	Render verification

4. Gate 5 Defense Presentation Roadmap & Team Roles

Presentation Segment	Speaker / Lead	Key Content & Demonstration Focus
1. Problem & Context (2m)	Blessing Edmund Kwame Dogbe	KNUST OPD overcrowding, manual triage bottlenecks, paper scheduling delays.
2. Solution & Product (3m)	Raymond B. Afrani	Dual-facility online booking, automated arrival windows, live corridor queue display.
3. Live Demonstration (6m)	Sterling Awuley & Harry Nartey	End-to-end flow: Mobile student booking, instant SMS, receptionist triage, corridor token, doctor room consultation.
4. Architecture & Security (2m)	Able Kafu Azanda & Emmanuella	Centralized RBAC matrix, student 2-booking cap, SMS OTP verification, fail-fast env validator, 391 tests.
5. Impact & Handover (2m)	Blessing Edmund Kwame Dogbe	65% peak congestion reduction, deployment runbooks, repository release v1.0.0, evaluator Q&A.

5. Definition of Done (DoD) Sign-Off Checklist

- Clinical Domain Logic:** Dual-facility booking, student 2-booking cap, atomic slot claim, past-slot rejection, arrival window, no-show slot release, SMS OTP cancellation.
- Security & Reliability:** Fail-fast startup env validator (env.js), centralized RBAC matrix (staffAuth.js), multi-tier rate limiting, phone masking, NoSQL operator sanitizer.
- Automated Quality:** 391 passing backend tests, 37 passing Playwright E2E browser tests, 0 build errors across frontend & prototype, clean ESLint.
- Deployment & Infrastructure:** Active Vercel frontend, active Render backend, MongoDB Atlas replica set with connection pooling (min 10 / max 50), mNotify SMS balance.
- Handover & Defense:** 15-minute slide deck rehearsed, 3-minute 1080p demo video archived, complete docs/ runbooks, GitHub release v1.0.0 tagged, AmaliTech defense presented.